

15.11.16 (E)

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

Nov 2016

Max. Marks: 100

Class: **S.Y. B. Tech**

Name of the Course: Applied Mathematics - III

Course Code : UCEC301 / UITEC301

Duration: 3hr.

Semester: III

Branch: Comp / IT

Instructions:

(1) All Questions are Compulsory

(2) Figures to right indicate full marks. Each sub-question has equal marks.

Q. No.	Questions		Marks
Q1.	A	Attempt any THREE of the following	18
	(i)	Find Laplace Transform of (i) $t\sqrt{1+\sin t}$ (ii) $\sin 2t \cos t \cosh 2t$	
	(ii)	Find the Inverse Laplace Transform of (i) $2\tanh^{-1}s$ (ii) $\frac{s^2-15s-11}{(s+1)(s-2)^2}$	
	(iii)	Find the Laplace Transformation of $f(t) = \begin{cases} \frac{t}{a} & 0 < t < a \\ \frac{1}{a}(2a-t) & a < t < 2a \end{cases}$ Where $f(t) = f(t+2a)$	
	(iv)	Using Laplace Transform evaluate $\int_0^\infty \frac{e^{-\sqrt{2}t}}{t} \sin t \sin ht dt$	
	B	Attempt any ONE of the following questions	07
	(i)	Using Convolution theorem find $L^{-1} \left[\frac{(s+2)^2}{(s^2+4s+8)^2} \right]$	
	(ii)	Solve $(D^2-D-2)y = 20 \sin 2t$ with $y(0) = 1$ & $y'(0) = 2$	
Q2.	A	Attempt any THREE of the following	18
	(i)	Find the Fourier series of $f(x) = \sqrt{1-\cos x}$ in $[0, 2\pi]$	
	(ii)	Find the half range sine Fourier series $f(x) = Lx - x^2$ in $[0, L]$	
	(iii)	Find the Fourier series of $f(x) = \cos x $ in $[-\pi, \pi]$	
	(iv)	Find complex form of Fourier series $f(x) = \cosh 3x + \sinh 3x$ in $[-3, 3]$	
	B	Attempt any ONE of the following questions	07
	(i)	Find the Fourier series $f(x) = x \sin x$ in $[-\pi, \pi]$. Hence deduce $\frac{\pi-1}{4} = \frac{1}{1 \cdot 3} - \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} - \dots$	
	(ii)	Find Fourier transform of $f(x) = \begin{cases} 1, & x < a \\ 0, & x > a \end{cases}$ hence find the value of $\int_0^\infty \frac{\sin x}{x} dx$	

Question No.	Questions		Max. Marks
Q3	A	Attempt any THREE of the following	18
	(i)	Prove that Eigen values of Hermitian Matrix are real. Hence Verify $A = \begin{bmatrix} 2 & 1+i \\ 1-i & 1 \end{bmatrix}$	
	(ii)	If $A = \begin{bmatrix} -1 & 4 \\ 2 & 1 \end{bmatrix}$ then prove that $3 \tan A = A \tan 3$	
	(iii)	Show that $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{bmatrix}$ is derogatory & find minimal polynomial.	
	(iv)	Find the Eigen values and Eigen vectors of A^3 where $A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$	
	B	Attempt any ONE of the following questions	7
	(i)	Show that $A = \begin{bmatrix} 8 & -8 & -2 \\ 4 & -3 & -2 \\ 3 & -4 & 1 \end{bmatrix}$ is diagonalisable. Find the transforming matrix and the diagonal matrix	
Q4	(ii)	Show that $A = \begin{bmatrix} 0 & c & -b \\ -c & 0 & a \\ b & -a & 0 \end{bmatrix}$ satisfying Cayley Hamilton Theorem and hence find A^{-1} if it exists.	
	A	Attempt any THREE of the following	18
	(i)	Find the work done in moving particle from A[1,0,1] to B[2,1,2] along the straight line AB in the force field $\vec{F} = x^2\hat{i} + (x-y)\hat{j} + (y+z)\hat{k}$	
	(ii)	Find the directional derivative $\phi = x^2 - y^2 + 2z^2$ at the point P[1,2,3] in the direction of the line PQ where Q is the point [5,0,4]. In what direction it will be maximum? Find the maximum value.	
	(iii)	Prove that $\nabla \cdot \left[r \nabla \left(\frac{1}{r^n} \right) \right] = \frac{n(n-2)}{r^{n+1}}$	
	(iv)	Evaluate $\oint [(x^2 + 2y)dx + (4x + y^2)dy]$ by Green's Theorem where C is the boundary of the region bounded by $y = 0$, $y = 2x$ and $x + y = 3$.	
	B	Attempt any ONE of the following questions	7
	(i)	Evaluate $\iiint \vec{F} \cdot d\vec{S}$ using divergence theorem where $\vec{F} = x^3\hat{i} + y^3\hat{j} + z^3\hat{k}$ and S is the surface of the sphere $x^2 + y^2 + z^2 = a^2$.	
	(ii)	Prove that the line integral of $\vec{F} = (y \sin z - \sin x)\hat{i} + (x \sin z + 2yz)\hat{j} + (xy \cos z + y^2)\hat{k}$ is independent of the path of integration and hence find the scalar potential ϕ	